PRACTICAL-5

Problem Definition:

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password,course, year, gender, and terms acceptance. Use Regular Expression for validations.

Key Questions / Analysis/ Interpretation

1. Are correct input types and attributes used?

Yes. Correct HTML5 input types and attributes like text, email, tel, password, radio, checkbox, id, and placeholder are used appropriately.

2. Are validation errors displayed near the relevant fields?

No. Currently, validation errors are displayed using alert pop-ups, not directly near the relevant fields**.**

3. Is password strength checked?

Yes. Password strength is checked using Regex for a minimum of 8 characters, including at least one letter and one number.

4. Is the form accessible using keyboard and screen-reader-friendly Labels?

Yes. The form can be accessed using the keyboard, and labels are provided for the input fields, making it screen-reader-friendly.

Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Student Registration Form</title>
   <link rel="stylesheet" href="style.css">
   <script src="script.js"></script>
</head>
<body>
   <div class="container">
      <h1>Student Registration Form</h1>
      <form id="registrationForm" onsubmit="return validateForm()">
         <label>Full Name:</label>
         <input type="text" id="name" placeholder="Enter your full name">
         <br><br>
         <label>Enrollment Number:</label>
         <input type="text" id="enrollment"placeholder="Example: 25DCE001">
         <br><br>
         <label>Email ID:</label>
         <input type="email" id="email"placeholder="example@charusat.edu.in">
         <br><br>
         <label>Mobile Number:</label>
         <input type="tel" id="mobile"placeholder="Enter 10 digit mobile number">
         <br><br>
         <label>Password:</label>
         <input type="password" id="password" placeholder="Enter password">
         <br><br>
         <label>Confirm Password:</label>
                  <input  type="password"  id="confirmPassword"placeholder="Re-enter
password">
         <br><br>
         <label>Course:</label>
         <select id="course">
            <option value="">Select Course</option>
            <option value="DCE"> Computer Engineering</option>
            <option value="IT"> Information Technology </option>
            <option value="CE"> Computer Science</option>
            <option value="EC">Electronics Engineering </option>
         </select>
         <br><br>
```

```
        <label>Year:</label>
        <select id="year">
            <option value="">Select Year</option>
            <option value="1">First Year</option>
            <option value="2">Second Year</option>
            <option value="3">Third Year</option>
            <option value="4">Fourth Year</option>
        </select>
        <br><br>
        <label>Gender:</label>
        <div class="gender">
            <input type="radio"name="gender" value="Male">
            <label>Male</label>
            <input type="radio"name="gender"value="Female">
            <label>Female</label>
            <input type="radio"name="gender" value="Other">
            <label>Other</label>
        </div>
        <br>
        <div class="terms">
            <input type="checkbox" id="terms">
            <label>agree to the Terms and Conditions
            </label>
        </div>
        <br>
        <button type="submit">Register </button>
        <button type="reset">Reset</button>
    </form>
  </div>
    <script src="script.js"></script>
</body>
</html>
```

Style.css:

```
body{
  font-family: Arial, sans-serif;
  background-color: lightblue;
}
.container{
  width:500px;
  margin:50px auto;
  padding:25px;
  background-color:white;
```

```
    border-radius:10px;
}
h1{
    text-align:center;
}
label{
    font-weight:bold;
}
input[type="text"],
input[type="email"],
input[type="tel"],
input[type="password"],
select{
    width:100%;
    padding:10px;
    margin-top:5px;
    box-sizing:border-box;
}
.gender{
    display:flex;
    align-items:center;
    gap:5px;
}
.gender label{
    margin-right:15px;
}
.terms{
    display:flex;
    align-items:center;
    gap: 5px;
}
.terms label{
    font-weight:normal;
}
button{
    padding:10px 20px;
    margin-right:10px;
    border:none;
    border-radius:5px;
    cursor:pointer;
}
```

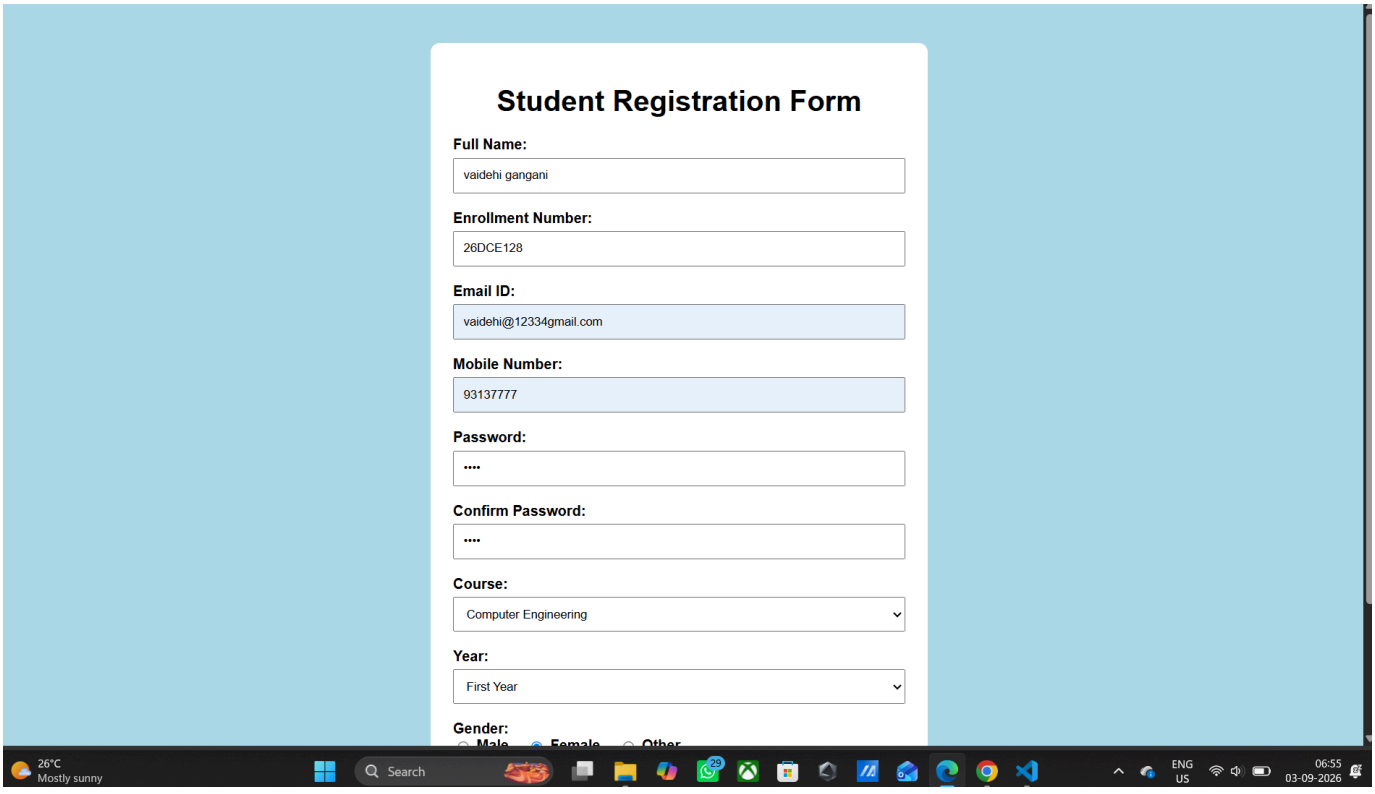
Script.js:

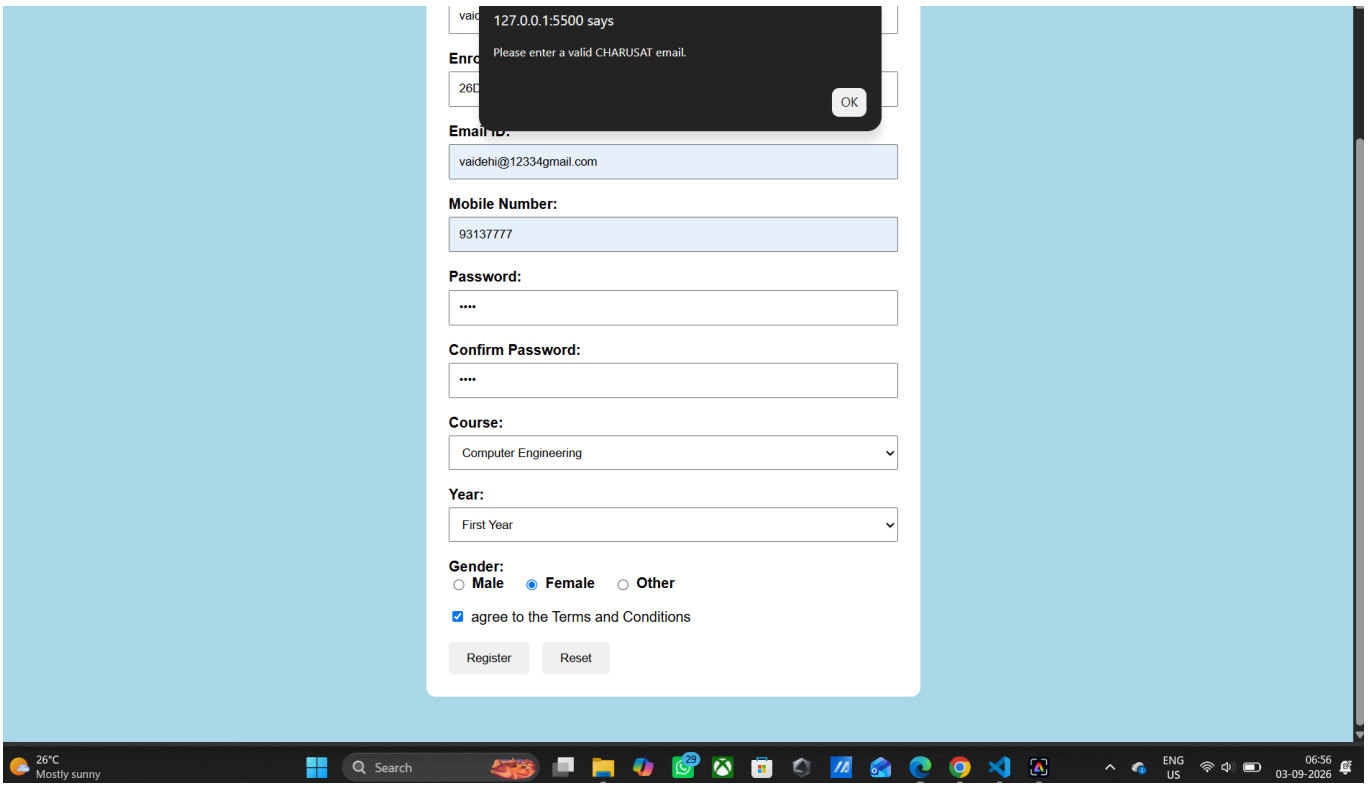
```
function validateForm()
{
    // Get values from form
    let name=document.getElementById("name").value;
    let enrollment=document.getElementById("enrollment").value;
    let email=document.getElementById("email").value;
    let mobile=document.getElementById("mobile").value;
    let password=document.getElementById("password").value;
    let confirmPassword=document.getElementById("confirmPassword").value;
    let course=document.getElementById("course").value;
    let year=document.getElementById("year").value;
    let terms=document.getElementById("terms").checked;

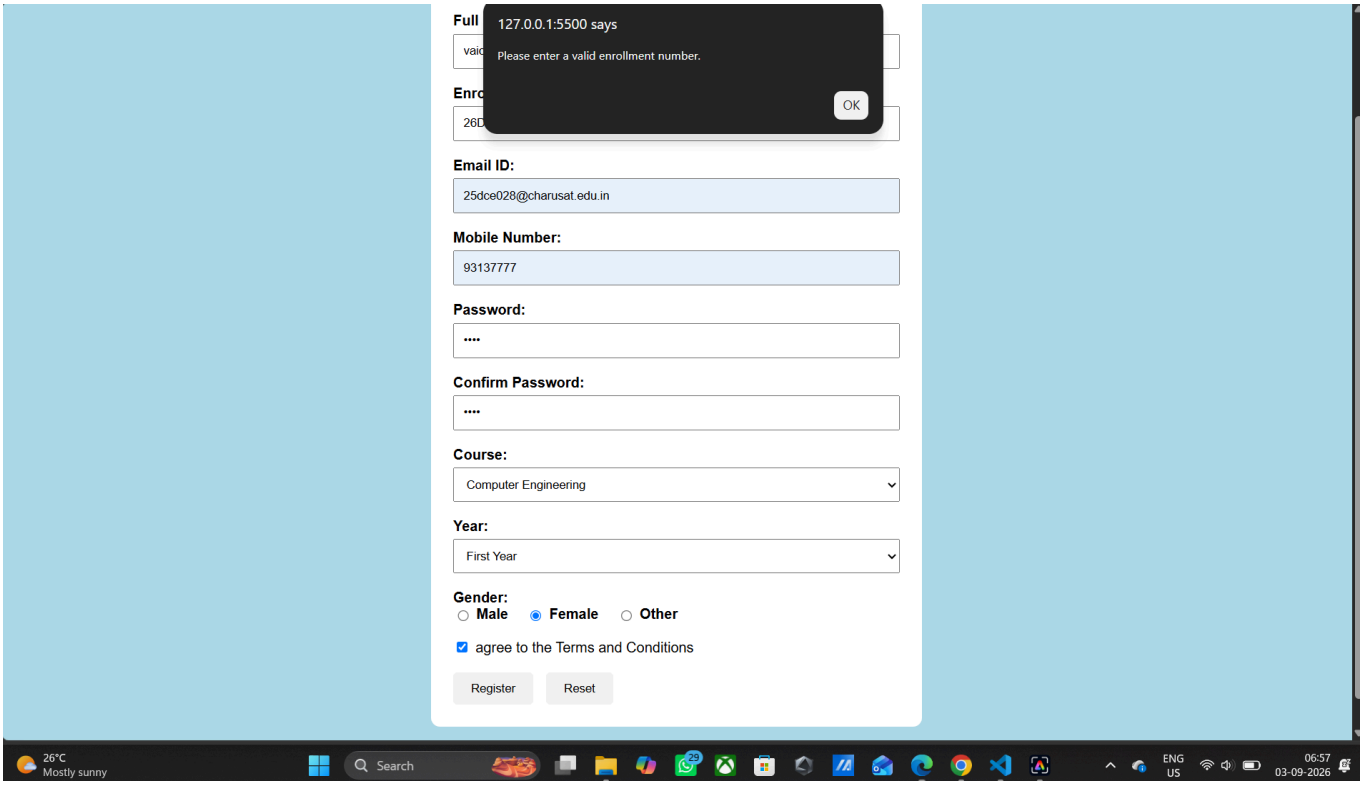
    // Regular Expressions
    let nameRegex = /^[A-Za-z ]{3,}$/;
    let enrollmentRegex = /^(25DCE[0-9]{3}|26DCE[0-9]{3})$/;
    let emailRegex = /^[A-Za-z0-9._%+-]+@charusat\.edu\.in$/;
    let mobileRegex = /^[6-9][0-9]{9}$/;
    let passwordRegex = /^(?=.*[A-Za-z])(?=.*[0-9]).{8,}$/;
    if (!nameRegex.test(name)){
        alert("Please enter a valid name.");
        return false;
    }
    if (!enrollmentRegex.test(enrollment)){
        alert("Please enter a valid enrollment number.");
        return false;
    }
    if (!emailRegex.test(email)){
        alert("Please enter a valid CHARUSAT email.");
        return false;
    }
console.log(mobile);
console.log(mobileRegex.test(mobile));
    if (!mobileRegex.test(mobile)){
        alert("Please enter a valid 10 digit mobile number.");
        return false;
    }
    if (!passwordRegex.test(password)){
            alert("Password must contain at least 8 characters, one letter and one
number.");
        return false;
```

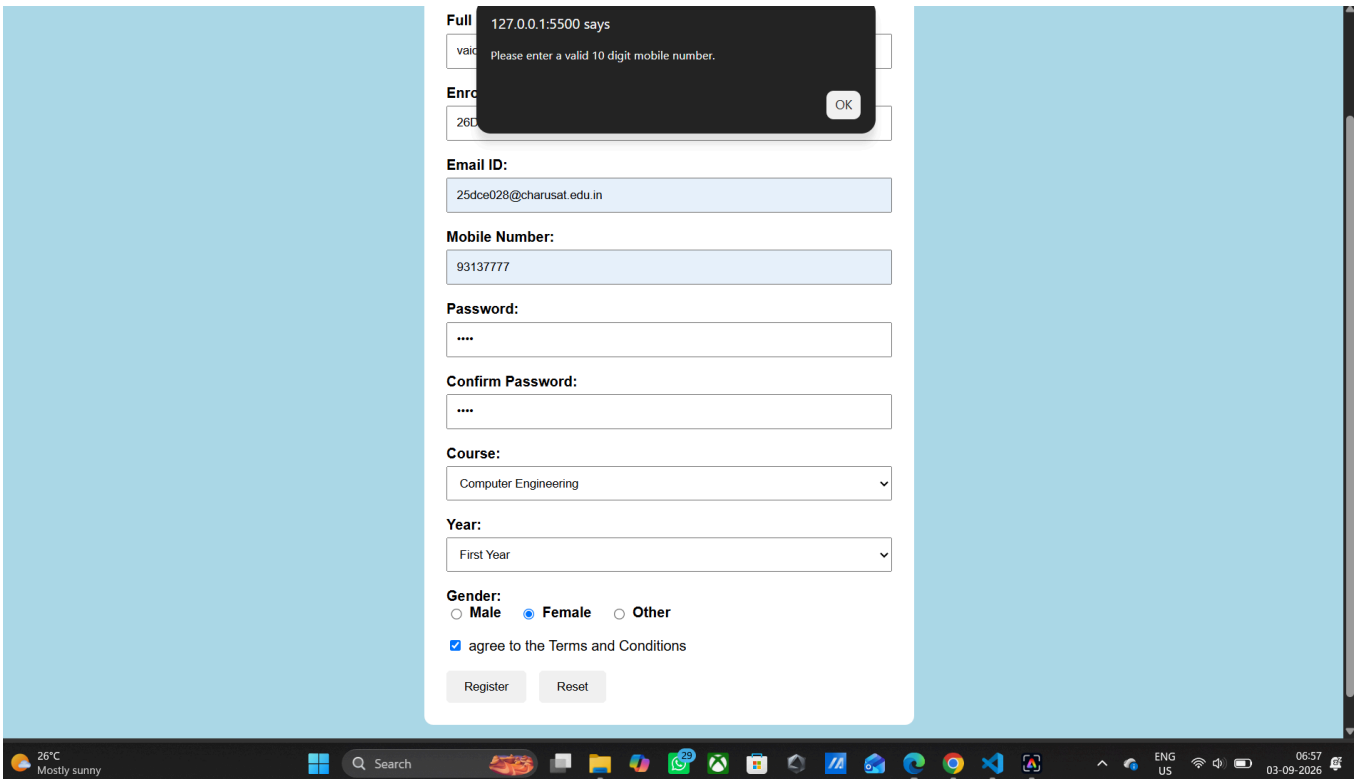
```
    }
    if (password != confirmPassword){
        alert("Password and Confirm Password do not match.");
        return false;
    }
    if (course == ""){
        alert("Please select a course.");
        return false;
    }
    if (year == ""){
        alert("Please select your year.");
        return false;
    }
    // Check Terms
    if (terms == false){
        alert("Please accept the Terms and Conditions.");
        return false;
    }
    alert("Registration Successful!");
    return true;
}
```

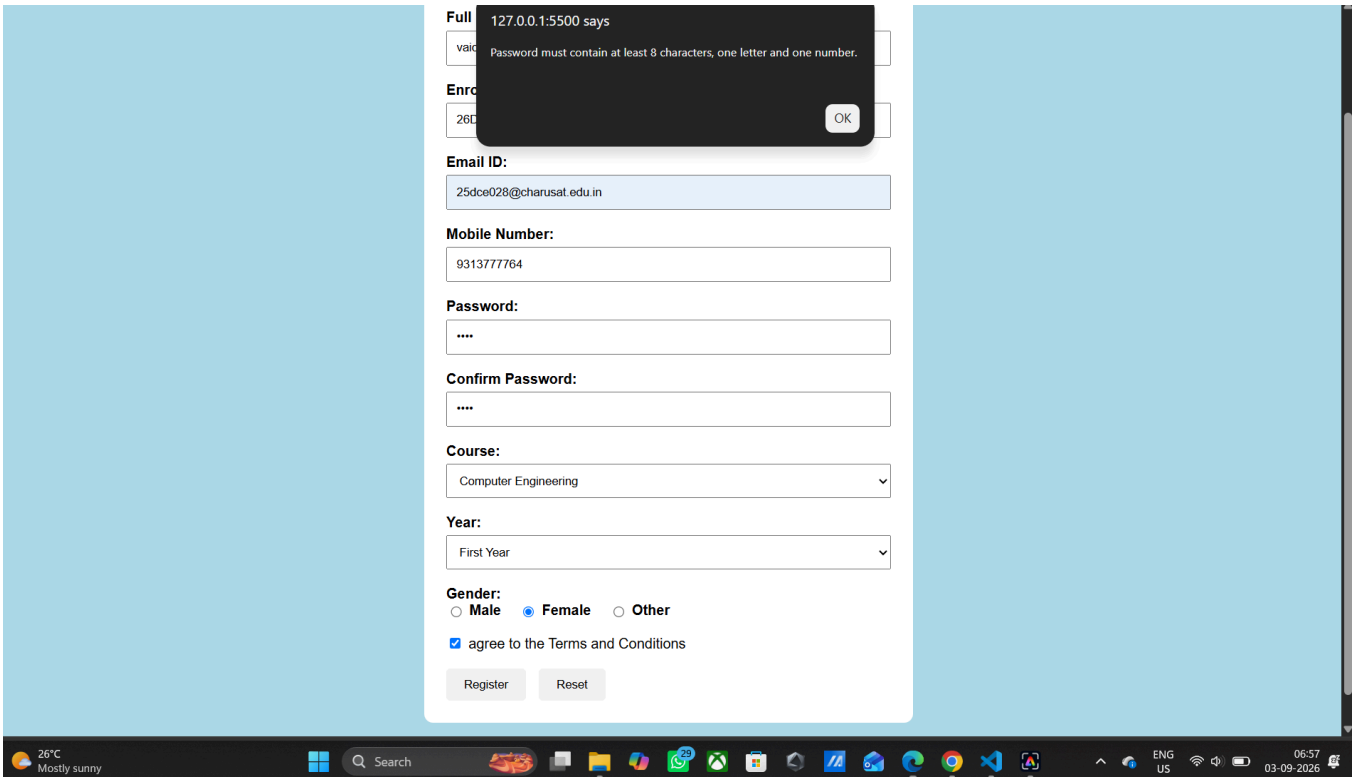
Output:

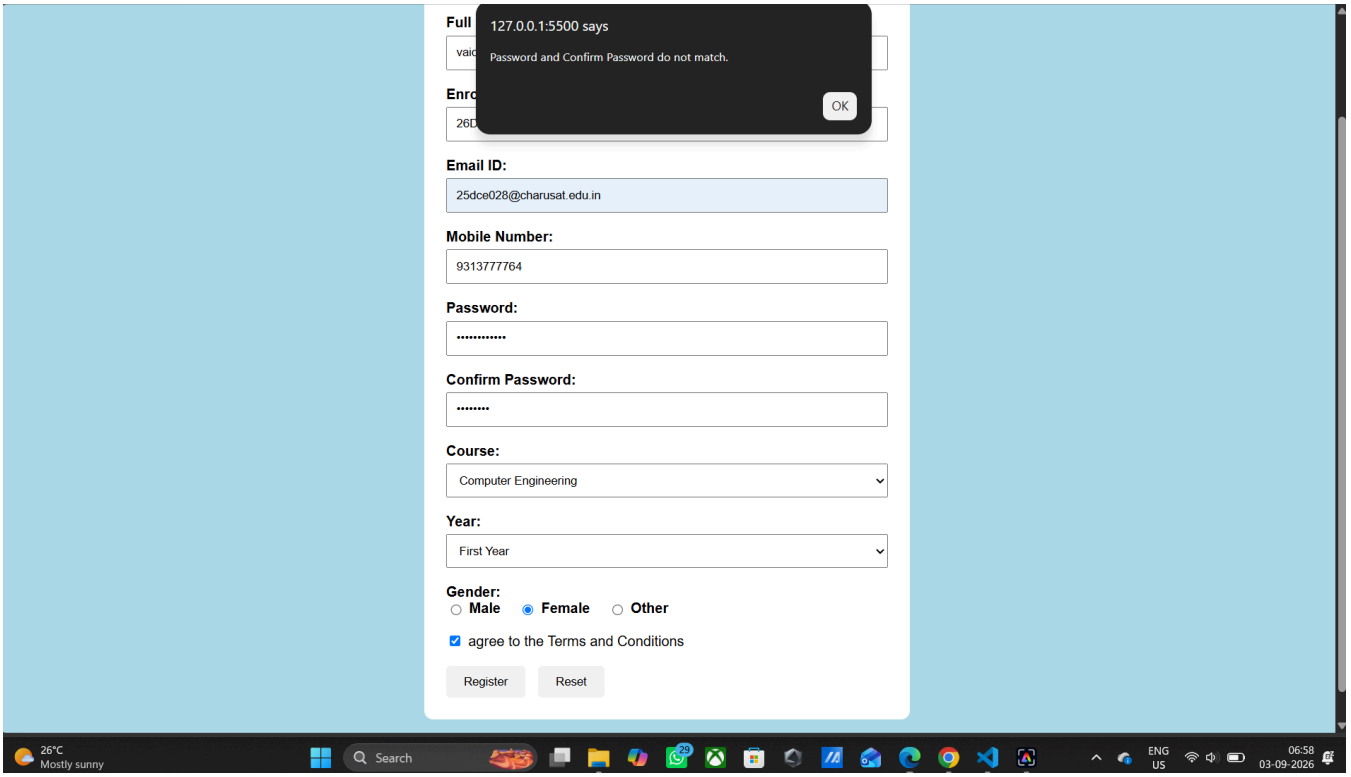


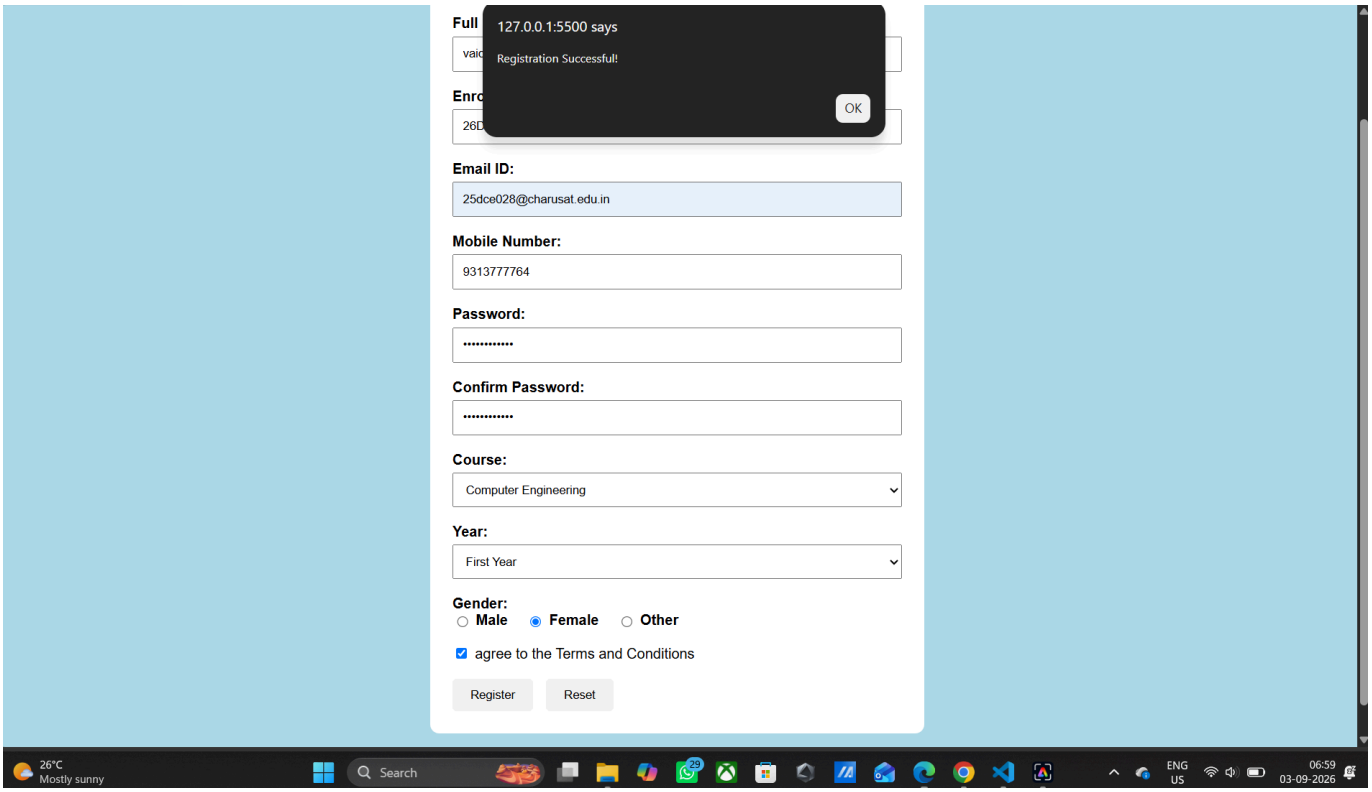

127.0.0.1:5500 says
Please enter a valid CHARUSAT email.
OK
Email ID:
vaidehi@12334gmail.com
Mobile Number:
93137777
Password:
Confirm Password:
Course:
Computer Engineering
Year:
First Year
Gender:
Male
Female
Other
agree to the Terms and Conditions
Register
Reset


127.0.0.1:5500 says
Please enter a valid enrollment number.
OK
Email ID:
25dce028@charusat.edu.in
Mobile Number:
93137777
Password:
Confirm Password:
Course:
Computer Engineering
Year:
First Year
Gender:
Male
Female
Other
agree to the Terms and Conditions
Register
Reset


127.0.0.1:5500 says
Please enter a valid 10 digit mobile number.
OK
Email ID:
25dce028@charusat.edu.in
Mobile Number:
93137777
Password:
Confirm Password:
Course:
Computer Engineering
Year:
First Year
Gender:
Male
Female
Other
agree to the Terms and Conditions
Register
Reset

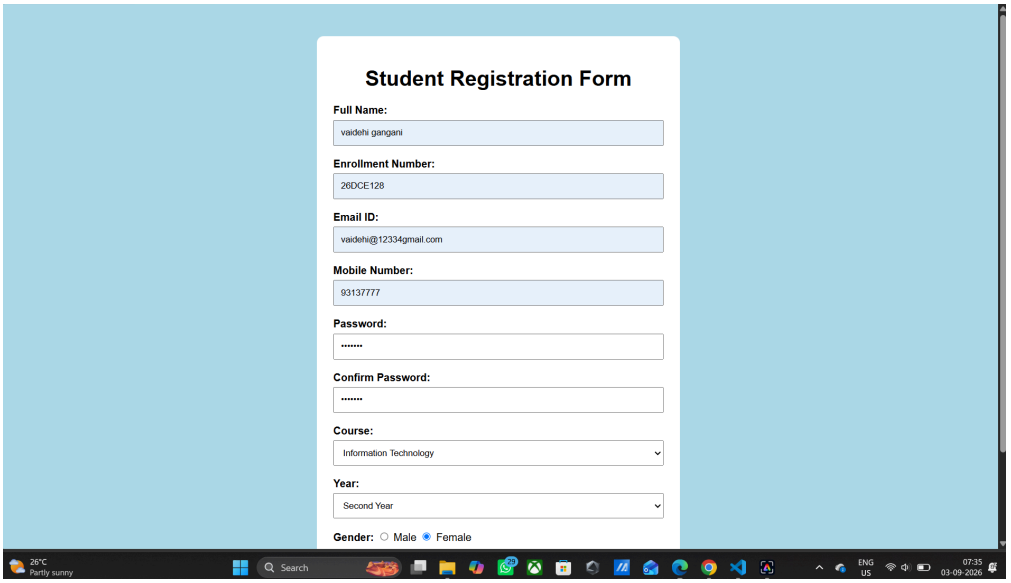

127.0.0.1:5500 says
Password must contain at least 8 characters, one letter and one number.
OK
Email ID:
25dce028@charusat.edu.in
Mobile Number:
9313777764
Password:
Confirm Password:
Course:
Computer Engineering
Year:
First Year
Gender:
Male
Female
Other
agree to the Terms and Conditions
Register
Reset


127.0.0.1:5500 says
Password and Confirm Password do not match.
OK
Email ID:
25dce028@charusat.edu.in
Mobile Number:
9313777764
Password:
Confirm Password:
Course:
Computer Engineering
Year:
First Year
Gender:
Male
Female
Other
agree to the Terms and Conditions
Register
Reset


127.0.0.1:5500 says
Registration Successful!
OK
Email ID:
25dce028@charusat.edu.in
Mobile Number:
9313777764
Password:
Confirm Password:
Course:
Computer Engineering
Year:
First Year
Gender:
Male
Female
Other
agree to the Terms and Conditions
Register
Reset

Challenges faced:

Challenge 1: Connecting HTML Form with JavaScript



Initially, the form was submitted directly without showing any validation message. The validateForm() function was not connected to the form. I solved this by adding onsubmit="return validateForm()" to the form.

Before:

<form id="registrationForm">

After:

<form id="registrationForm" onsubmit="return validateForm()">

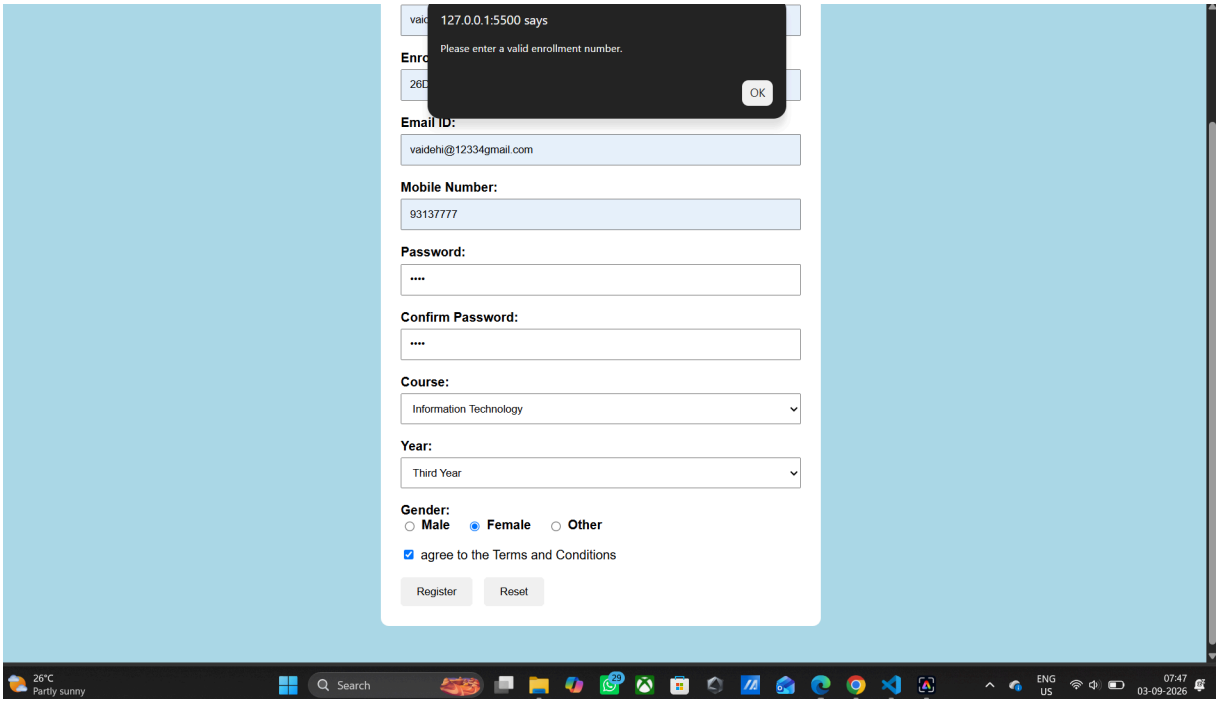
Challenge 2:

Initially, the Regex accepted only `25DCE001` format. I modified it to accept both `25DCE001` and `26DCE001` formats.

Before:

/^(25DCE[0-9]{3})$/

After:
/^(25DCE[0-9]{3}|26DCE[0-9]{3})$/



Challenge 3: Mobile Number Validation

Before:

let mobileRegex = /^[6-9][0-9]$/;

After:

let mobileRegex = /^[6-9][0-9]{9}$/;